QQ-C-523
JUNE 29, 1955

SUPERSEDING
Int. Fed. Spec. QQ-B-00731b (NAVY-SHIPS)
30 November 1953, and
Fed. Spec. QQ-B-731b
May 30, 1945

FEDERAL SPECIFICATION

COPPER ALLOY INGOTS; BRASS (YELLOW, HIGH STRENGTH) MANGANESE AND MANGANESE-ALUMINUM BRONZE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers high strength yellow brass (manganese and manganese-aluminum bronze) ingots for remelting. (See 6.1.)

1.2 Classification. — The ingots shall be of the following alloys, as specified (see 6.1):

Alloy A.
Alloy B.
Alloy C.
Alloy D.
Alloy E.
Alloy F.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN-B-621—Boxes, Wood, Nailed and Locked-Corner.
QQ-M-151—Metals, General Specification for Inspection of.
PPP-B-601—Boxes, Wood, Cleated-Plywood.

Military Specifications:

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN-P-132—Packaging and Packing for Overseas Shipment—Crates, Unsheathed, Wood; Nailed (for Maximum Net Load of 2,500 Pounds).

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 Ingots shall conform to the chemical composition as shown in table I.

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TABLE I.—*Chemical composition*

Alloy	Copper	Aluminum	Manganese	Iron	Tin	Lead	Nickel (maximum)	Zinc
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*
A	55 – 60	0.5 – 1.5	1.50[1]	0.4 – 2.0	1.00[1, 2]	0.30[1]	0.50	Remainder
B and C ..	60 – 68	3.0 – 7.5	2.5 – 5.0	2.0 – 4.0	0.10[1]	0.10[1]	1.0	Remainder
D	56 – 62	0.25 – 1.0	0.1 – 0.5	0.75 – 1.5	0.5 – 1.0	0.5 – 1.0	0.50	Remainder
E	66 – 68	4.5 – 6.0	3.0 – 5.0	2.0 – 4.0	0.10[1]	0.10[1]	1.0	Remainder
F	56 – 61	0.7 – 1.0	1.50[1]	0.4 – 2.0	1.00[1, 2]	0.30[1]	0.50	38 – 40

[1] Maximum.

[2] A minimum of 0.50 percent tin shall be specified (see 6.1) when the purpose for which manganese bronze is to be used requires maximum resistance to dezincification.

3.2 Mechanical properties.

3.2.1 Ingots, when remelted and cast in sand, shall conform to the properties shown in table II.

TABLE II.—*Mechanical properties*

Alloy	Tensile strength (minimum)	Yield[1] strength (minimum)	Elongation in 2 inches (minimum)
	Psi.	*Psi.*	*Percent*
A	65,000	.....	20
B	90,000	45,000	18
C	110,000	60,000	12
D	60,000	20,000	15
E	90,000	45,000	18
F	65,000	.....	20

[1] 0.5 percent extension under load or 2.2 percent offset.

3.3 Weights. — Unless otherwise specified (see 6.1), ingots shall be furnished in commercial weights.

3.4 Marking for identification.—The manufacturer's name or brand and the alloy shall be cast or stamped on each ingot. Each ingot shall bear its proper heat or melt number either stamped on or cast in, so that its identity will be clear at the point of delivery.

3.5 Workmanship.—Ingots shall be of uniform quality, clean and free from excessive dirt, dross, slag inclusions, surface exudations, burned or gassed metal and shall not exhibit a "puffed" condition.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—For purpose of sampling, a lot shall consist of not more than 100,000 pounds of ingots from a single melt or not more than 2,000 pounds of ingots poured from a continuous melting furnace.

4.1.2 *Sampling for inspection.*—For the purpose of inspection specified in 4.2, the Government inspector shall select representative sample ingots from each lot as shown in table III.

TABLE III.—*Sampling for inspection*

Number of ingots in lot	Number of ingots in sample
Under 12	All
12 – 100	11
Over 100	12

4.1.3 *Sampling for tests.* — At least 4 ingots shall be selected by the Government inspector from each lot for tests.

4.1.3.1 *Chemical analysis.*—Samples from 2 of the ingots selected in accordance with 4.1.3 shall be obtained by drilling entirely through 2 segments of each ingot (one at the end and one centrally located). Drillings shall include metal at both surfaces. Drillings shall not be mixed, except that drillings taken from one ingot may

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be thoroughly mixed to make one sample. Each sample shall consist of not less than 2 ounces.

4.1.3.2 *Tension tests.*

4.1.3.2.1 After the samples for chemical analysis have been taken, at least two test coupons shall be cast in sand from the ingots selected in accordance with 4.1.3. When practicable, the Government inspector shall witness the pouring of the test bars, and their removal from the mold. He shall take such action as necessary to identify the castings and test bars with the heat or lot number assigned and shall stamp each test bar for future identification with the proper heat or lot number and suitable identification stamp.

4.1.3.2.2 When it is not practicable for the Government inspector to witness the pouring of the test coupons he shall require the manufacturer to establish and maintain procedures and records that will assure the inspector that test coupons are poured, processed, and identified as to the proper heat, melt or lot number of castings they represent.

4.1.3.2.3 When the Government inspector has reasonable doubt as to the identity of the test coupons, drillings for chemical analysis shall be taken from the coupons and the castings purported to be from the same lot for check analysis, to assist the inspector in establishing identity. If the results of any check analysis are conclusive that the test coupons do not represent the lot involved, the Government inspector shall take proper steps to have the manufacturer correct his procedures and maintain proper identity of test bars.

4.1.3.2.4 Unless otherwise specified in the contract or order, the chilling of test coupons shall be cause for rejection of the material represented.

4.1.3.2.5 The test coupon shall conform to the form and dimensions shown on figure 1. Other test bar designs may be used if approved by the bureau or agency concerned. The specimen machined therefrom shall conform to the form and dimensions shown for type I of Federal Specification QQ–M–151.

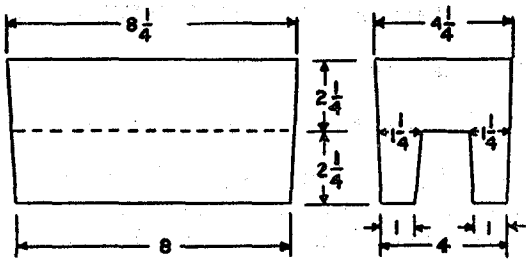


FIGURE 1.—Test coupon.

4.2 Inspection.—Each ingot selected in accordance with 4.1.2 shall be inspected by the Government inspector to determine conformance with this specification. If one or more sample ingots fail to conform to this specification, the entire lot shall be rejected. A rejected lot may be resubmitted, after the manufacturer has reinspected each ingot in the lot for the deficiency noted, and has removed all nonconforming ingots.

4.3 Test procedures.

4.3.1 *Chemical analysis.*—Each of the samples obtained in accordance with 4.1.3.1 shall be forwarded to a Government laboratory designated by the procuring activity or a laboratory satisfactory to the Government inspector for chemical analysis to determine conformance with 3.1.1. If either of the samples fail to conform to this specification, the lot shall be rejected.

4.3.2 *Tension tests.* — One of the test specimens obtained in accordance with 4.1.3.2 shall be pulled in tension in accordance with Federal Specification QQ–M–151 to determine conformance with 3.2. If the specimen fails to conform to this specification, except as provided in 4.3.2.1, the lot shall be rejected.

4.3.2.1 If any test specimen contains flaws which result in the failure of the specimen, an additional specimen from the same melt may be selected to replace it at the discretion of the Government inspector. The nature of the flaw or defect in the discarded specimen, when deemed characteristic of the material under inspection shall be taken into consideration when determining the acceptability of the material.

4.3.3 *Retests.*—Retests for mechanical properties will be permitted in accordance with Federal Specification QQ–M–151.

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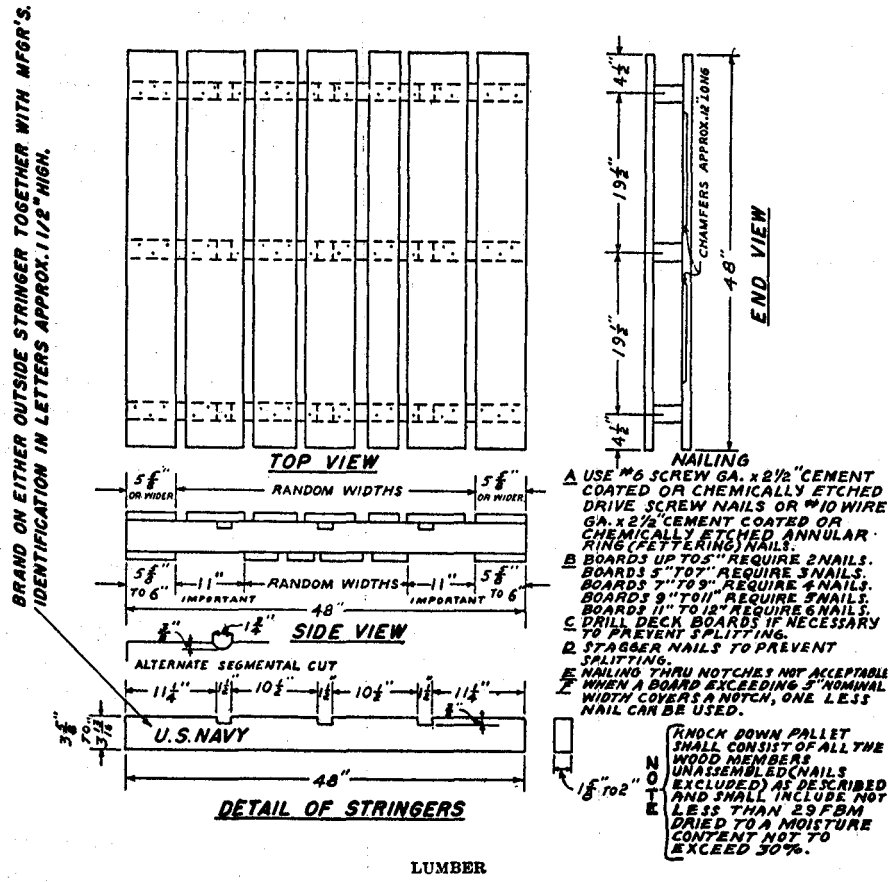
5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *For less than carload shipments.*

5.1.1.1 *For domestic shipment and storage.*—Unless otherwise specified in the contract or order, the subject commodity shall be packed in snug fitting wood-cleated plywood or nailed wood boxes, or in nailed wood crates, conforming to Federal Specification PPP–B–601 (domestic type), NN–B–621, or Military Specification JAN–P–132, respectively. The gross weight of wood boxes shall not exceed 500 pounds, and crates shall not exceed 1000 pounds.

5.1.1.2 *For overseas shipment.*—When specified (see 6.1), the subject commodity shall be packed in wood-cleated plywood or nailed wood boxes or unsheathed nailed wood crates conforming to Federal Specification PPP–B–601 (overseas type), Military Specifications JAN–P–106, JAN–P–132, respectively. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of wood boxes shall



A—Use sound square edge lumber with no defects that will materially impair strength of piece contemplating its use in a finished pallet.

B—Use following woods; any stage of seasoning: white ash, beech, birch, rock elm, hackberry, hickory, hard maple, oak, pecan.

C—Use random width boards 3" or wider, surfaced one side ⅞" bit or miss. Only one full 3" board allowed per surface. Surfaced faces to be exposed faces.

D—Space between top deck boards to be 1" min. and 1½" max. space between bottom deck boards not to exceed 1½" except 11" spaces as noted.

E—Boards must be placed so that each notch is completely covered by a single board of not less than 4" nominal width.

F—Length tolerance plus or minus ¼".

FIGURE 2.—48- by 48-inch standard hardwood pallet (assembled).

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not exceed 500 pounds, and unsheathed crates shall not exceed 1000 pounds.

5.1.1.3 *For domestic or overseas shipment.*— When practicable, containers shall contain an identical number of ingots of one alloy, and shall be of uniform size. In addition, all containers shall withstand storage, rehandling, and reshipment without the necessity of repacking.

5.1.2 *For carload shipments.*

5.1.2.1 *For domestic shipment.* — Unless otherwise specified in the contract or order, ingots, when shipped in carload lots, shall be palletized and securely strapped with not less than ¾- by 0.035-inch strapping. The gross pallet load shall not exceed approximately 3,000 pounds. Pallets shall be standard pallets 48 by 48 inches, constructed in conformance with figure 2 or equivalent.

5.2 Marking.

5.2.1 *For nonmilitary agencies.* — Marking for shipment shall be as specified in the contract or order.

5.2.2 *For military agencies.*—In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Alloy required (see 1.2).
(c) Weights of the ingots, if not to be furnished in commercial weights (see 3.3).
(d) When a minimum of 0.50 percent tin is required (see table I).
(e) Whether preparation for domestic or overseas shipment is required (see 5.1.1.1 and 5.1.1.2).
(f) Marking required. (See 5.2.1.)

6.2 Quality control. — The requirements specified in 4.1.1 for a lot to consist of a single melt may be modified upon application by the contractor or prospective contractor to the procuring activity, via the Government inspector, giving detailed information that the contractor or prospective contractor has demonstrated his ability to produce ingots of uniform quality in accordance with this specification.

6.3 The packing and marking specified herein apply to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

6.4 Federal Specifications do not include all types, classes, grades, and sizes, of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, classes, grades, and sizes which are suitable for Federal Government requirements.

6.5 Ingots covered by this specification are equivalent to castings of the following compositions:

Alloy A	MIL–B–16443 (class 1)	QQ–B–726 (class A)
Alloy B	MIL–B–16522 (class 2)	QQ–B–726 (class B)
Alloy C	MIL–B–16522 (class 1)	QQ–B–726 (class C)
Alloy D	.....	QQ–B–726 (class D)
Alloy E	MIL–B–16522 (class 3)	
Alloy F	(modification of Alloy A)	

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—O Q T
Navy—Sh A Or
Air Force.

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